■ LabInsight AI

Medical Laboratory Data Analysis & Clinical Decision-Support System

Generated: 2026-10-10 23:19
Report ID: #1
Confidential Medical Record

Patient Identification & Report Details

Patient ID:	P001	**Report Date:**	2026-10-10 23:19:59
Patient Name:	Test Patient	**Age:**	45 years
Gender:	Female	**Contact:**	9876543210

AI Disease Risk Predictions

Target Pathology	Model Prediction	Model Confidence	Model Architecture
■ **Diabetes Mellitus**	**Diabetes Risk Detected**	88.5%	Logistic Regression Pipeline (Imputer + Scaler)
■ **Chronic Kidney Disease**	**No Kidney Disease Risk Detected**	95.0%	ColumnTransformer Pipeline (OneHot + Scaler + LR)
❤■ **Heart Disease**	**Heart Disease Risk Detected**	72.3%	Logistic Regression Pipeline (StandardScaler + LR)

Laboratory Test Findings & Parameter Breakdown

Test Parameter	Patient Value	Standard Reference Range	Diagnostic Flag
Glucose (Fasting/Random)	**140 mg/dL**	70.0 - 99.0 mg/dL	**High**
Blood Pressure (Systolic)	**130 mmHg**	90.0 - 120.0 mmHg	**High**
Serum Cholesterol	**220 mg/dL**	125.0 - 200.0 mg/dL	**High**
Serum Creatinine	**1.10 mg/dL**	0.6 - 1.2 mg/dL	**Normal**

Explainable AI (SHAP) — Diabetes Feature Attribution

SHAP (SHapley Additive exPlanations) isolates the mathematical contribution of each laboratory parameter to the diabetes classification model. Positive impacts increase predicted risk; negative impacts lower predicted risk.

Parameter	SHAP Impact	Effect Direction	Contribution Level
Blood Glucose (mg/dL)	**+2.1500**	Increases model output	High
Body Mass Index (kg/m²)	**+0.4500**	Increases model output	Moderate

■■ ***MEDICAL & ACADEMIC DISCLAIMER:*** *LabInsight AI is an educational research and clinical decision-support tool. The predictions and SHAP explainability analyses presented in this document are generated by machine learning models and statistical pipelines for academic evaluation. They do NOT constitute a medical diagnosis, clinical prognosis, or treatment directive. All findings must be reviewed, verified, and interpreted by a licensed medical practitioner.*